Q13 Comprehension Check

Question: A colleague proposes a new knowledge-editing system: "We'll find the single neuron that stores each fact — say, the Eiffel Tower's location — and rewrite its outgoing weights. Superposition doesn't matter because we'll operate at the neuron level." Identify the two key mistakes in this plan, referencing (a) the causal-tracing evidence about where facts are mediated and (b) the superposition account of what individual neurons contain.

Answer: (1) Facts are not mediated by single neurons: causal tracing (ROME, Meng et al. 2022) localizes factual recall to middle-layer *MLP modules* processing the last subject token — a distributed, module-level site, not an individual neuron — and the successful edits (ROME/MEMIT) are rank-one *matrix* updates in a linear-associative-memory view, not neuron rewrites. (2) Superposition (Elhage et al. 2022) means individual neurons are polysemantic: each participates in many feature directions, and features span more directions than there are neurons. Editing "at the neuron level" would therefore co-edit every unrelated feature sharing that neuron — bleedover by construction. The colleague has the granularity backwards: the safe level is directions/matrices with covariance-regularized updates, not neurons.